ENTERPRISE PROCUREMENT & CANVASS ARCHITECTURE

PartsCompare Enterprise Sourcing Platform

System Architecture, Governance & Multi-Role Operational Standard Operating Procedure (SOP)

SYSTEM STATUS	DATABASE SYNC	STATUTORY TAX	SECURITY & AUTH
v2.4 Enterprise Production	Local-First + Supabase Cloud	BIR 12% VAT (RA No. 9337)	RBAC + 2FA PIN + Audit Ledger

1. Executive Overview & Platform Capabilities

PartsCompare is a production-grade enterprise procurement and sourcing governance platform tailored for commercial mechanical and HVAC engineering contractors (modeled around *AMGC Airconditioning Services* and *Nexus CoolTech Industrial Philippines*). The application eliminates manual spreadsheet bid tabulation, guarantees compliance with Philippine VAT laws, and enforces strict corporate governance through immutable audit trails.

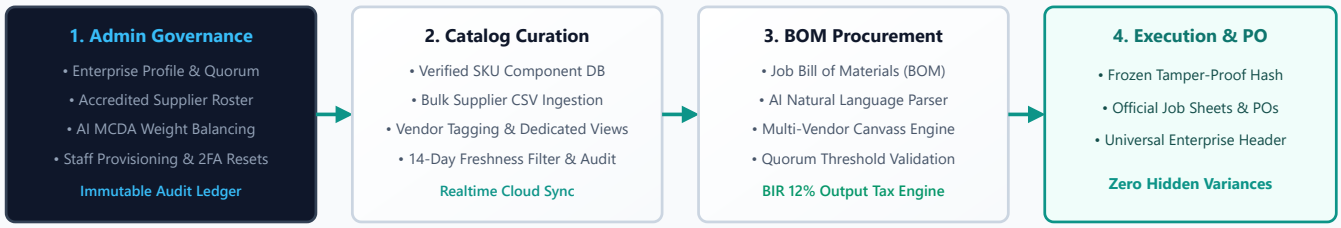


FIGURE 1.1: PARTSCOMPARE END-TO-END ENTERPRISE PROCUREMENT LIFECYCLE

Core System Enhancements Implemented

🏢 Enterprise Canvass Quorum & Supplier Directory

Removed rigid 3-vendor limits. Corporations can register unlimited vendors (4, 5, 10+) and configure custom **Required Supplier Quotes** thresholds. The comparison engine verifies whether each part meets required canvass bids before PO issuance.

📦 Dedicated Catalog Attribution & Filtering

Offers are directly tagged to accredited suppliers. The Catalog Workspace provides a **Filter by Supplier** selector, and the Admin Directory includes direct "📦 **View Catalog**" shortcuts for instant vendor audits.

🛡️ Identity-Verified Anti-Impostor Security

Replaced email dependencies with a secure **6-Digit Security PIN** and **Unique Security Question**. Features 2-factor Admin self-recovery, staff approval queues, and **5-attempt lockout** against brute-force attacks.

PH Official Philippine BIR 12% VAT Engine

Full statutory compliance with Republic Act No. 9337. Features dual viewing modes: Net of VAT (Vatable Sales), 12% Output Tax, and Gross Invoice Payable. Incomplete line items are protected against ₱0.00 zero-filling.

⚡ Supabase Cloud PostgreSQL & Realtime Sync

Local-first architecture backed by cloud PostgreSQL with 8 synchronized tables and WebSocket broadcast channels for seamless multi-device updates across field technicians and procurement officers.

🤖 AI Assisted BOM Parser & MCDA Optimizer

Converts unstructured technician job notes directly into structured BOM line items. The multi-criteria decision analysis (MCDA) engine balances cost, stock surplus, freshness, and single-supplier delivery efficiency.

2. Role-Based Access Control (RBAC) Matrix

Enterprise Role	Personnel Profile	Primary Operational Scope	Accessible Workspaces

ADMINISTRATOR
karen_admin
	Operations Manager, General Manager, Lead Auditor	Company profile branding, statutory tax rates, canvass quorum rules, supplier accreditation/suspension, MCDA algorithm weights, user provisioning, and audit logs.	`#/` (Admin Console)
			`#/catalog`, `#/lists`
			`#/summaries`

CATALOG MAINTAINER **pearl_dba**	Procurement Officer, Inventory Controller, Purchasing Agent	Curating verified component SKUs, uploading bulk supplier quotation spreadsheets (CSV), filtering vendor inventories, and tracking 14-day quote freshness.	`#/catalog` (Offer Catalog) `#/faqs`, `#/about`

BOM PREPARER
dave_pm
	Field HVAC Technician, Project Estimator, Site Engineer	Generating job Bills of Materials (BOM), parsing field notes via AI, comparing live quotes against canvass quorum, balancing 12% VAT, and freezing job sheets.	`#/lists` (BOM Workspace)
			`#/compare/:id`
			`#/summaries`

3. Operational Role Operating Guides (SOP)

Role 1: Enterprise Administrator (karen_admin)

Responsible for corporate policy enforcement, supplier accreditation, AI weight tuning, and staff account governance.

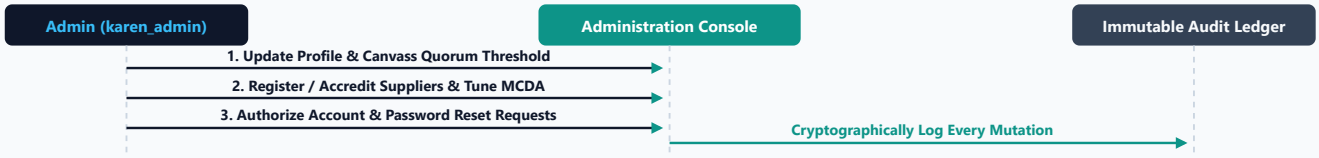


FIGURE 3.1: ENTERPRISE ADMINISTRATOR GOVERNANCE LIFECYCLE

Step 1.1: Configure Enterprise Profile & Canvass Quorum

1. Sign in as `karen_admin`. The system routes you directly to the **Enterprise Administration Console** (`#/`).
2. In the **Enterprise Business Profile & Universal Branding** card, configure:
   - **Business Name, Industry, and Headquarters Address:** Dynamically displayed in headers, footers, and job sheets.
   - **Max Quote Age Threshold:** Defaults to `14` days. Quotes older than this are flagged as stale.
   - **Required Supplier Quotes (Canvass Quorum):** Set to `3`, `4`, or `5` bids per tender policy.
3. Click **Save Enterprise Business Profile**. All modules instantly adopt the updated settings.

Step 1.2: Manage Accredited Supplier Network

1. Scroll down to the **Accredited Supplier Network & Vendor Directory** section.
2. Click `+ Add Accredited Supplier` to enroll a new partner with Trade Name, Contact Email, and Phone.
3. Click `📦 View Catalog (X)` on any supplier row to immediately filter the Offer Catalog to that specific vendor.
4. To restrict a vendor with quality or warranty issues, click `Suspend`. Suspended suppliers are excluded from BOM quotes.

Step 1.3: Tune AI MCDA Procurement Weights

1. In the **AI Procurement Engine Settings (MCDA Utility Weights)** panel, adjust the 4 decision criteria:
   - **Lowest Unit Cost Weight (%):** Prioritizes immediate cash purchase savings.
   - **Stock Buffer / Surplus Ratio (%):** Favors suppliers with sufficient inventory to avoid split deliveries.
   - **Quote Freshness / Recency (%):** Rewards quotes updated within the last 48–72 hours.
   - **Supplier Multi-line Consolidation (%):** Groups items under a single accredited vendor to cut freight costs.
2. Ensure the total adds up to exactly **100%** (live badge: `Total: 100% ✓ Balanced`). Click **Save MCDA Weights**.

Step 1.4: Authorize Staff Account & Password Reset Requests

1. Review incoming requests under **Staff Account Access & Provisioning Requests**. Verify the name and requested role, then click `✓ Approve & Activate`.
2. Review verified staff requests under **Identity-Verified Password Reset Requests** and click `✓ Approve Reset`.
3. Inspect the **Immutable Audit Event Ledger** at the bottom of the page to review tamper-evident logs for every event.

Role 2: Catalog Maintainer (pearl_dba)

Responsible for maintaining accurate, fresh, and accredited supplier pricing for field component procurement.



FIGURE 3.2: CATALOG MAINTAINER INGESTION & FRESHNESS WORKFLOW

Step 2.1: Audit Existing Catalog Quotations

1. Sign in as `pearl_dba`. You will be placed directly in the **Offer Catalog Workspace** (`#/catalog`).
2. Review top metric cards: **Total Offers**, **Fresh & Complete**, **Stale (>14d)**, and **Suppliers Tracked**.
3. Use the `Filter by Supplier:` selector to inspect pricing from a single partner (e.g., *Everwell Parts*, *C&C Airconditioning Supply*, or *Manco Refrigeration*).
4. Click the **Stale** filter chip to instantly isolate quotes requiring updated price bulletins from vendors.

Step 2.2: Upload a Bulk Supplier Catalog via CSV

1. Click `CSV Template` to download the standard `supplier_catalog_template.csv`.
2. Populate the spreadsheet with the mandatory vendor schema:

Column Header	Format	Example Value	Description
`part_id`	Text (Exact SKU)	`CBB65-45UF-450V`	Standard engineering part number
`description`	Text	`45uF 450V Motor Run Capacitor`	Component description & specs
`source`	Text	`Everwell Parts Inc.`	Accredited supplier name
`attributed_source`	Text	`Quote #EW-2026-4491`	Vendor quote / PO invoice ref
`unit_price`	Decimal (PHP)	`380.00`	Unit purchase price
`stock`	Integer	`45`	Current warehouse stock inventory

3. Click `Upload Supplier CSV` and select your file. The system validates all lines, automatically links them to the accredited vendor, marks them fresh, and synchronizes to Supabase Cloud.

Step 2.3: Add or Edit an Individual Component Offer

1. Click `+ Add New Offer`.
2. Select the **Exact Part ID** and the designated **Accredited Supplier** from the dropdowns.
3. Input Unit Price, Available Quantity, Checked Date, and Invoice Reference. Click **Save Offer**.

💡 **Vendor Catalog Independence:** Multiple accredited suppliers can provide offers for the identical SKU. The system tracks each vendor's price, stock, and quotation date independently, allowing the canvass engine to perform true side-by-side market audits.

Role 3: BOM Preparer / Field Technician (dave_pm)

Responsible for building accurate job BOMs, running multi-supplier canvass analysis, and freezing PO job sheets.

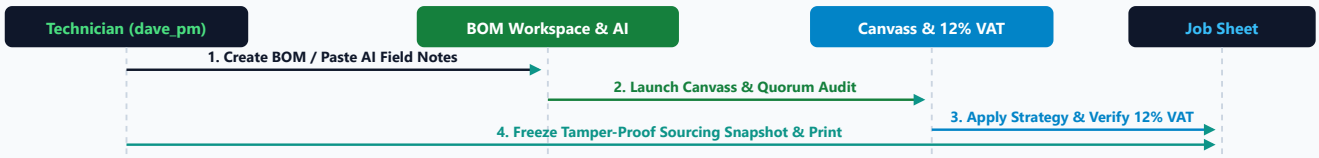


FIGURE 3.3: BOM PREPARER MULTI-VENDOR SOURCING & EXECUTION SOP

Step 3.1: Create a Job Bill of Materials (BOM)

1. Sign in as `dave_pm`. You land in the **Preparer Workspace** (`#/lists`).
2. Under **Create New BOM List**, enter the Job Reference (e.g., `Chiller Plant Overhaul Job #CH-2026-014`) and click `+ Create Blank BOM List` .
3. Add parts line-by-line using the **Exact Part ID** selector and specify required job quantities.
4. *AI Quick-Import Option:* Click `AI CSV Import` , paste informal field notes (e.g., `45uf capacitor, 2 / defrost contactor, 1 / r410a freon 1kg, 2`), click **Run AI Parser**, and click **Replace / Import Parts**.
5. Click **Save BOM Changes**.

Step 3.2: Run Multi-Supplier Comparison & Quorum Verification

1. Click **Compare Saved BOM** (or open `#/compare/:id`).
2. The comparison matrix displays competing supplier quotes side-by-side:
   - **Canvass Quorum Compliance Badge:** Evaluates whether each item has met the required number of competitive quotes (e.g., `3 / 3 Required Quotes ✓` or ⚠️ **Below 4-quote quorum**).
   - **Candidate Selection:** Radio buttons allow manual selection of individual vendor quotes.
3. Use the **AI Sourcing Strategy Optimizer** to auto-assign choices:
   - `Lowest Cost` : Minimizes purchase cost.
   - `Single Supplier` : Consolidates under a single vendor to reduce multi-vendor logistics.
   - `Stock Resilience` : Favors vendors holding ample stock buffer to protect against delivery stockouts.
   - `Balanced MCDA` : Optimizes according to the enterprise weights established by the Administrator.

Step 3.3: Verify Philippine BIR 12% VAT Breakdown

1. Toggle between VAT modes:
   - **Dual Mode (Standard):** Simultaneously displays VAT-Inclusive Gross Invoice, Net of VAT (Vatable Sales), and 12% Output Tax.
   - **Inclusive Mode:** Displays total gross payable.
   - **Exclusive Mode:** Displays net pre-tax base cost.
2. Incomplete or unquoted lines are explicitly marked and never zero-filled, preventing understated job bids.

Step 3.4: Freeze and Print Sourcing Summary

1. When all components are resolved, click `Freeze & Save Sourcing Summary` .
2. The snapshot is sealed with a cryptographic hash and stored immutably against subsequent vendor price modifications.
3. Click `Print Job Sheet` to produce the formal job sheet complete with corporate branding, tax breakdown, and vendor order details.

4. Identity-Verified Password Recovery SOP

Staff Recovery Workflow (pearl_dba & dave_pm)

1. On the Sign In page (`#/login`), click **Forgot password?**.
2. Enter your registered username and click **Verify Account & Continue**.
3. The system challenges you with your registered **Unique Security Question** and requests your **6-Digit Security PIN**.
4. Upon inputting correct credentials, your identity is cryptographically validated and queued in the **Admin Reset Queue**.
5. The Administrator clicks **Approve Reset** in the Admin Console, immediately committing your new password.

Administrator 2FA Self-Recovery (karen_admin)

1. Administrators cannot rely on a higher supervisor for password resets.
2. The recovery modal challenges the Administrator with strict **Two-Factor Authentication (2FA)**.
3. The Administrator must correctly enter **BOTH** the 6-Digit PIN *and* the exact Security Question answer.
4. Upon passing dual verification, the Administrator password updates immediately without delays.
5. An emergency audit entry (`ADMIN_2FA_PASSWORD_RECOVERY`) is logged to the immutable ledger.

🛡️ **Anti-Impostor Brute-Force Lockout Defense:** If any unauthorized user enters 5 consecutive incorrect PIN or security question guesses, the account is locked for **5 minutes**. A `SECURITY_ACCOUNT_LOCKOUT` alert is dispatched to the audit log with the client timestamp.

5. Default Demonstration & Operational Credentials

Role Name	Username	Demo Password	6-Digit PIN	Security Question & Verified Answer
ENTERPRISE ADMINISTRATOR	`karen_admin`	`demo`	`890415`	What city was your company founded in? → **manila**
CATALOG MAINTAINER	`pearl_dba`	`demo`	`350922`	What is your primary warehouse location? → **quezon city**
BOM PREPARER (TECHNICIAN)	`dave_pm`	`demo`	`240811`	What city was your primary high school located? → **cainta**

6. Supabase Cloud Schema & Storage Architecture

Table Name	Storage	Realtime	Data Description & Role Access
`company_settings`	PostgreSQL	Yes	Corporate profile, universal branding, currency, quote age threshold, canvass quorum.
`suppliers`	PostgreSQL	Yes	Accredited supplier network, contact credentials, accreditation status (active/suspended).
`catalog_offers`	PostgreSQL	Yes	Component pricing catalog, attributed supplier ID, unit prices, warehouse stock, checked date.
`bom_items`	PostgreSQL	Yes	Technician job BOM line items, required quantities, targeted engineering SKUs.
`sourcing_summaries`	PostgreSQL	Yes	Frozen procurement snapshots, selected quotes, 12% VAT calculations, cryptographic fingerprint.
`account_requests`	PostgreSQL	Yes	Staff access registration queue, requested role, encrypted initial credentials.
`password_reset_requests`	PostgreSQL	Yes	Identity-verified password reset requests awaiting Administrator authorization.
`audit_logs`	PostgreSQL	Yes	Append-only, immutable event log capturing all administrative mutations and security alerts.

> 💡 **Operational Readiness:** All demonstration credentials above are pre-seeded in the database fixtures. Personnel can update their 6-Digit PIN and Secret Question anytime via the `🛡️ Recovery PIN` button in the navigation bar.